

Recap on 1-D and 2-D Kinematics

SS ILP Physics (Intermediate), Week 2 Class 1



Outline

- Recap on 1-D and 2-D Kinematics
- Practice questions

Kinematics (in x-direction)

Position: $\vec{r}(t) = x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = \frac{dx}{dt}$$

$$v_x > 0, \text{direction} + \hat{i}$$

$$v_x < 0, \text{direction} - \hat{i}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

$$a_x = \frac{dv_x}{dt}$$

$$a_x > 0, \text{direction} + \hat{i}$$

$$a_x < 0, \text{direction} - \hat{i}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x(t_1) = v_{x,0} + \int_{t_0}^{t_1} a_x(t) dt$$

$$v_x > 0, \text{direction} + \hat{i}$$

$$v_x < 0, \text{direction} - \hat{i}$$

Position: $\vec{r}(t) = x \hat{i}$

$$x(t_1) = x_0 + \int_{t_0}^{t_1} v_x(t) dt$$

$$x > 0, \text{direction} + \hat{i}$$

$$x < 0, \text{direction} - \hat{i}$$

Kinematics (in y-direction)

Position: $\vec{r}(t) = y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = \frac{dy}{dt}$$

$$v_y > 0, \text{ direction } + \hat{j}$$

$$v_y < 0, \text{ direction } - \hat{j}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

$$a_y = \frac{dv_y}{dt}$$

$$a_y > 0, \text{ direction } + \hat{j}$$

$$a_y < 0, \text{ direction } - \hat{j}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y(t_1) = v_{y,0} + \int_{t_0}^{t_1} a_y(t) dt$$

$$v_y > 0, \text{ direction } + \hat{j}$$

$$v_y < 0, \text{ direction } - \hat{j}$$

Position: $\vec{r}(t) = y \hat{j}$

$$y(t_1) = y_0 + \int_{t_0}^{t_1} v_y(t) dt$$

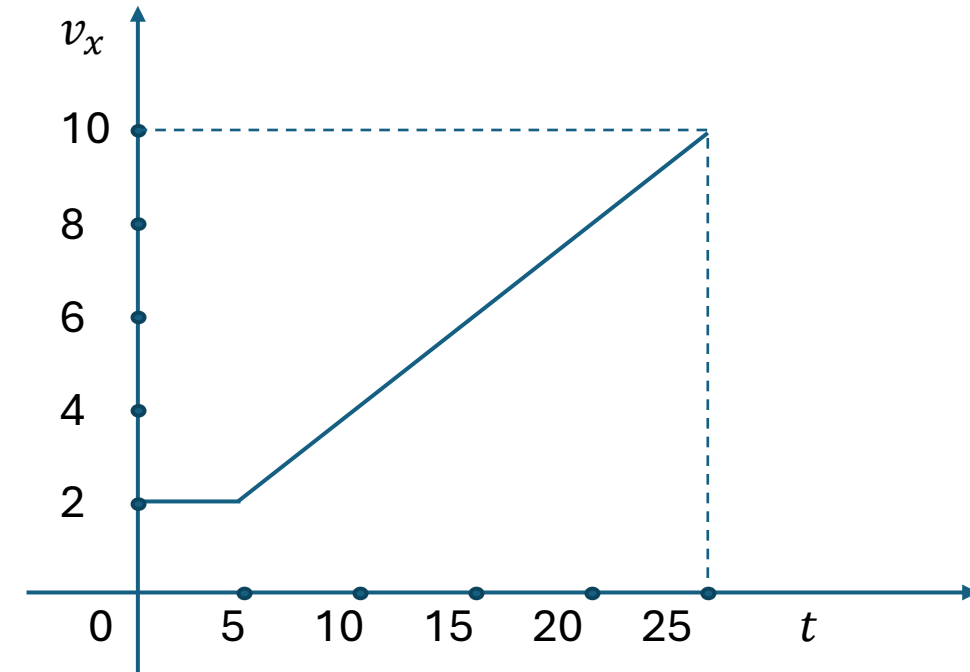
$$y > 0, \text{ direction } + \hat{j}$$

$$y < 0, \text{ direction } - \hat{j}$$

Case Problem 1

The diagram shows a velocity (m/s) vs time (s) graph for a particle moving along a straight line.

1. What is the acceleration of the particle between 0s and 5s?
2. What is the acceleration of the particle between 5s and 25s?



<https://www.chegg.com/homework-help/questions-and-answers/question-3-diagram-shows-velocity-time-graph-particle-moving-along-straight-line-determine-q85895811>

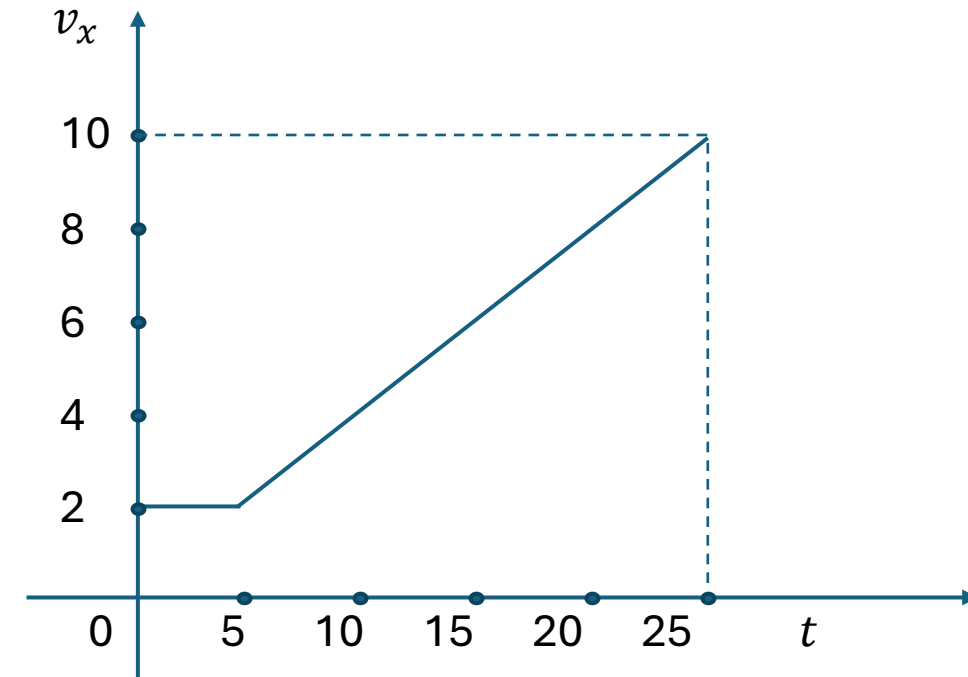
Case problem 1: Solution

The diagram shows a velocity (m/s) vs time (s) graph for a particle moving along a straight line in x direction.

1. What is the acceleration of the particle between 0s and 5s?
2. What is the acceleration of the particle between 5s and 25s?

1. $a_x = \frac{dv_x}{dt} = 0 \text{ m/s}^2$
2. $a_x = \frac{dv_x}{dt} = 0.4 \text{ m/s}^2$

<https://www.chegg.com/homework-help/questions-and-answers/question-3-diagram-shows-velocity-time-graph-particle-moving-along-straight-line-determine-q85895811>



Recall: Constant acceleration

$$v_x(t) = v_{x,0} + a_x t$$

$$x(t) = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2$$

- For constant acceleration, acceleration along the x -axis = a_x

- Velocity, $v_x(t) = v_{x,0} + \int_0^t a_x dt = v_{x,0} + a_x t$

For your information only!

- Position, $x(t) = x_0 + \int_0^t v_x(t) dt = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2$

- Note: these equations are only valid for constant acceleration

<https://openstax.org/books/university-physics-volume-1/pages/3-6-finding-velocity-and-displacement-from-acceleration>

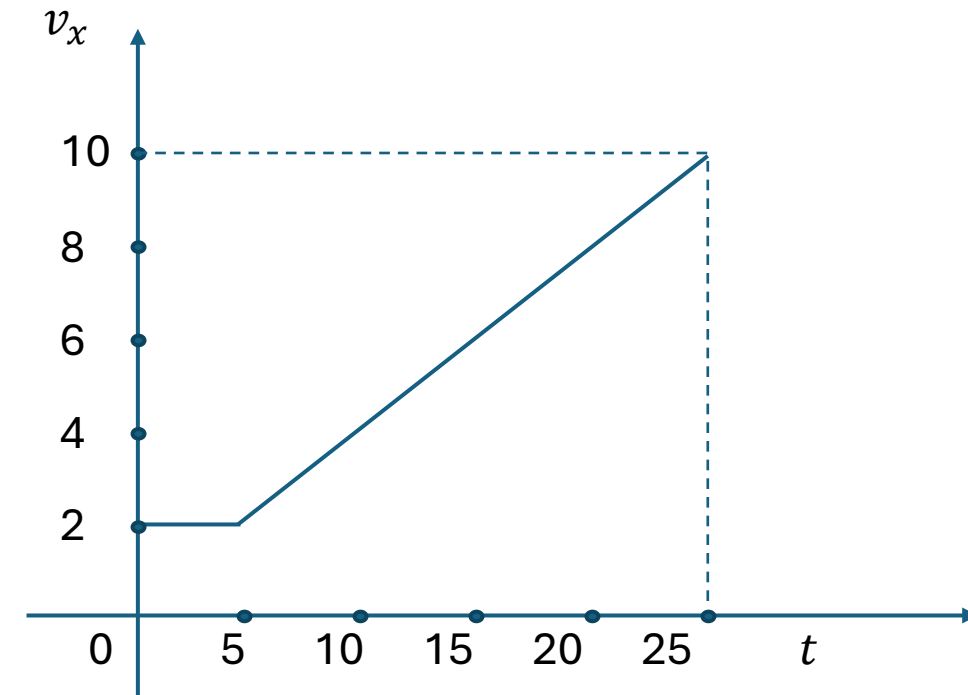
Case Problem 2

The diagram shows a velocity (m/s) vs time (s) graph for a particle moving along a straight line.

What is the displacement of the particle between 0s and 25s?

- A. 90m
- B. 120m
- C. 130m
- D. 150m

<https://www.chegg.com/homework-help/questions-and-answers/question-3-diagram-shows-velocity-time-graph-particle-moving-along-straight-line-determine-q85895811>



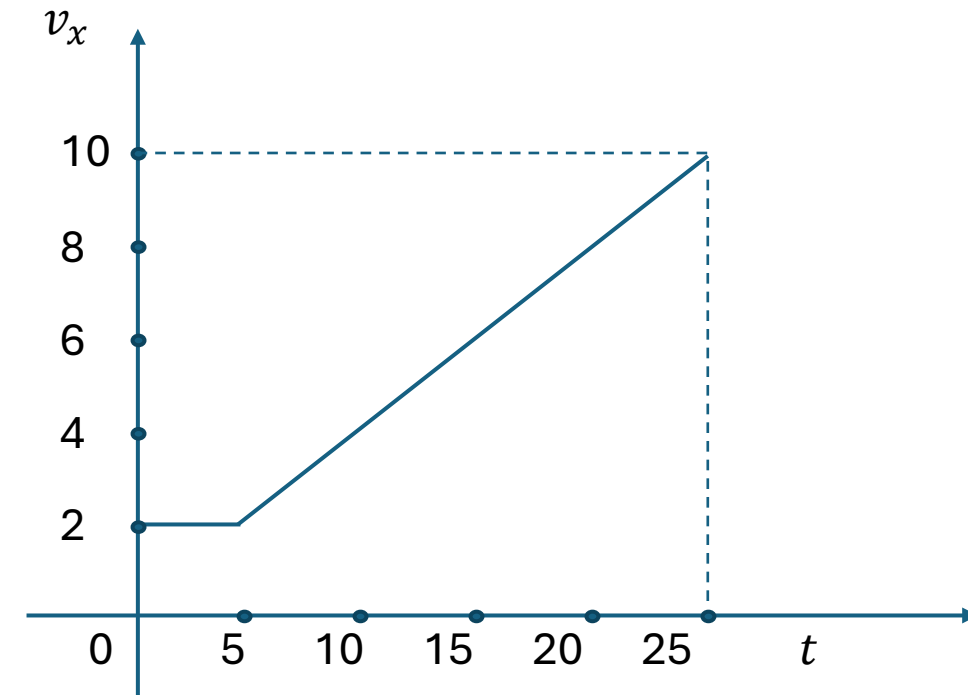
Case Problem 2: Solution

The diagram shows a velocity (m/s) vs time (s) graph for a particle moving along a straight line.

What is the displacement of the particle between 0s and 25s?

Ans: C

$$\begin{aligned}\text{Displacement} &= \text{area under the curve} \\ &= 25(2) + 0.5(10-2)(25-5) \\ &= 130 \text{ m}\end{aligned}$$



Kinematics (in x-direction): Restricted constant acceleration

Position: $\vec{r}(t) = x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = \frac{dx}{dt} \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

$$a_x = \frac{dv_x}{dt} \quad \begin{array}{l} a_x > 0, \text{ direction } + \hat{i} \\ a_x < 0, \text{ direction } - \hat{i} \end{array}$$

Acceleration: $\vec{a}(t) = a_x \hat{i}$

Velocity: $\vec{v}(t) = v_x \hat{i}$

$$v_x = v_{x,0} + a_x t \quad \begin{array}{l} v_x > 0, \text{ direction } + \hat{i} \\ v_x < 0, \text{ direction } - \hat{i} \end{array}$$

Position: $\vec{r}(t) = x \hat{i}$

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 \quad \begin{array}{l} x > 0, \text{ direction } + \hat{i} \\ x < 0, \text{ direction } - \hat{i} \end{array}$$

$$v_x^2 - v_{x,0}^2 = 2a_x(x - x_0)$$

Kinematics (in y-direction): Restricted constant acceleration

Position: $\vec{r}(t) = y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = \frac{dy}{dt} \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

$$a_y = \frac{dv_y}{dt} \quad \begin{array}{l} a_y > 0, \text{ direction } + \hat{j} \\ a_y < 0, \text{ direction } - \hat{j} \end{array}$$

Acceleration: $\vec{a}(t) = a_y \hat{j}$

Velocity: $\vec{v}(t) = v_y \hat{j}$

$$v_y = v_{y,0} + a_y t \quad \begin{array}{l} v_y > 0, \text{ direction } + \hat{j} \\ v_y < 0, \text{ direction } - \hat{j} \end{array}$$

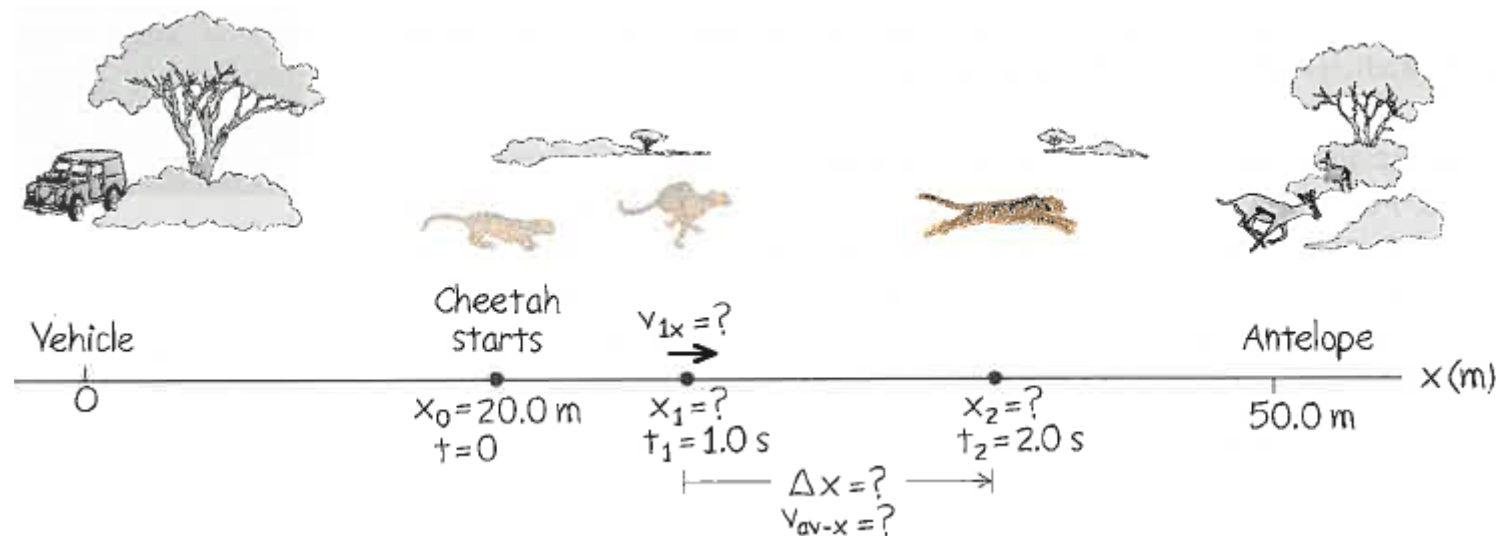
Position: $\vec{r}(t) = y \hat{j}$

$$y = y_0 + v_{y,0}t + \frac{1}{2}a_y t^2 \quad \begin{array}{l} y > 0, \text{ direction } + \hat{j} \\ y < 0, \text{ direction } - \hat{j} \end{array}$$

$$v_y^2 - v_{y,0}^2 = 2a_y(y - y_0)$$

Case Problem 3:

A cheetah is crouched 20 m to the east of a vehicle. At time $t=0$ the cheetah begins to run due east toward an antelope that is 50m to the east of the vehicle. During the first 2.0 s of the chase, the cheetah is accelerating from rest with $a = 10.0 \text{ m/s}^2$. (a) Find the cheetah's displacement between $t_1 = 1.0 \text{ s}$ and $t_2 = 2.0 \text{ s}$. (b) Find its average velocity during that interval. (c) Find its instantaneous velocity at $t_1 = 1.0 \text{ s}$.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.1, p62

Case Problem 3: Solution

A cheetah is crouched 20 m to the east of a vehicle. At time $t=0$ the cheetah begins to run due east toward an antelope that is 50m to the east of the vehicle. During the first 2.0 s of the chase, the cheetah is accelerating from rest with $a = 10.0 \text{ m/s}^2$. (a) Find the cheetah's displacement between $t_1 = 1.0 \text{ s}$ and $t_2 = 2.0 \text{ s}$. (b) Find its average velocity during that interval. (c) Find its instantaneous velocity at $t_1 = 1.0 \text{ s}$.

$$x = x_0 + v_{x,0}t + \frac{1}{2}a_x t^2 = 20 + 5t^2 \rightarrow x(1) = 20 + 5 \times 1^2 = 25 \text{ m}$$

$$x(2) = 20 + 5 \times 2^2 = 40 \text{ m}$$

$$\Delta x = x(2) - x(1) = 15 \text{ m}$$

$$v_{avg,x} = \frac{\Delta x}{\Delta t} = \frac{x(2) - x(1)}{2 - 1} = 15 \text{ m/s}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 2.1, p62

Case Problem 3: Solution

A cheetah is crouched 20 m to the east of a vehicle. At time $t=0$ the cheetah begins to run due east toward an antelope that is 50m to the east of the vehicle. During the first 2.0 s of the chase, the cheetah is accelerating from rest with $a = 10.0 \text{ m/s}^2$. (a) Find the cheetah's displacement between $t_1 = 1.0 \text{ s}$ and $t_2 = 2.0 \text{ s}$. (b) Find its average velocity during that interval. (c) Find its instantaneous velocity at $t_1 = 1.0 \text{ s}$.

$$v_x = v_{x,0} + a_x t = 10t \quad \rightarrow \quad v_x(1) = 10 \times 1 = 10 \text{ m/s}$$

Kinematics (Projectile motion)

Position: $\vec{r}(t) = x \hat{i} + y \hat{j}$

Velocity: $\vec{v}(t) = v_x \hat{i} + v_y \hat{j}$

$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt}$$

Acceleration:

$$\vec{a}(t) = a_x \hat{i} + a_y \hat{j} = -g \hat{j}$$

$$a_x = 0$$

$$a_y = -g$$

Acceleration: $\vec{a}(t) = a_y \hat{j} = -g \hat{j}$

Velocity: $\vec{v}(t) = v_x \hat{i} + v_y \hat{j}$

$$v_x(t) = v_{x,0}$$

$$v_y(t) = v_{y,0} - gt$$

Position: $\vec{r}(t) = x \hat{i} + y \hat{j}$

$$x = x_0 + v_{x,0}t$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$v_y^2 - v_{y,0}^2 = -2g(y - y_0)$$

Case Problem 4

A projectile is launched from the origin with an initial speed of 20.0 m/s at an angle of 35.0° above the horizontal. Take $g = 9.81 \text{ m/s}^2$. Find the velocity of the projectile at time $t = 0.5 \text{ s}$.

Case Problem 4: Solution

$$v_{x,0} = v_0 \cos \theta = 20 \times \cos 35.0^\circ = 16.38 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 20 \times \sin 35.0^\circ = 11.47 \text{ m/s}$$

$$v_x(t) = v_{x,0} = 16.38 \text{ m/s}$$

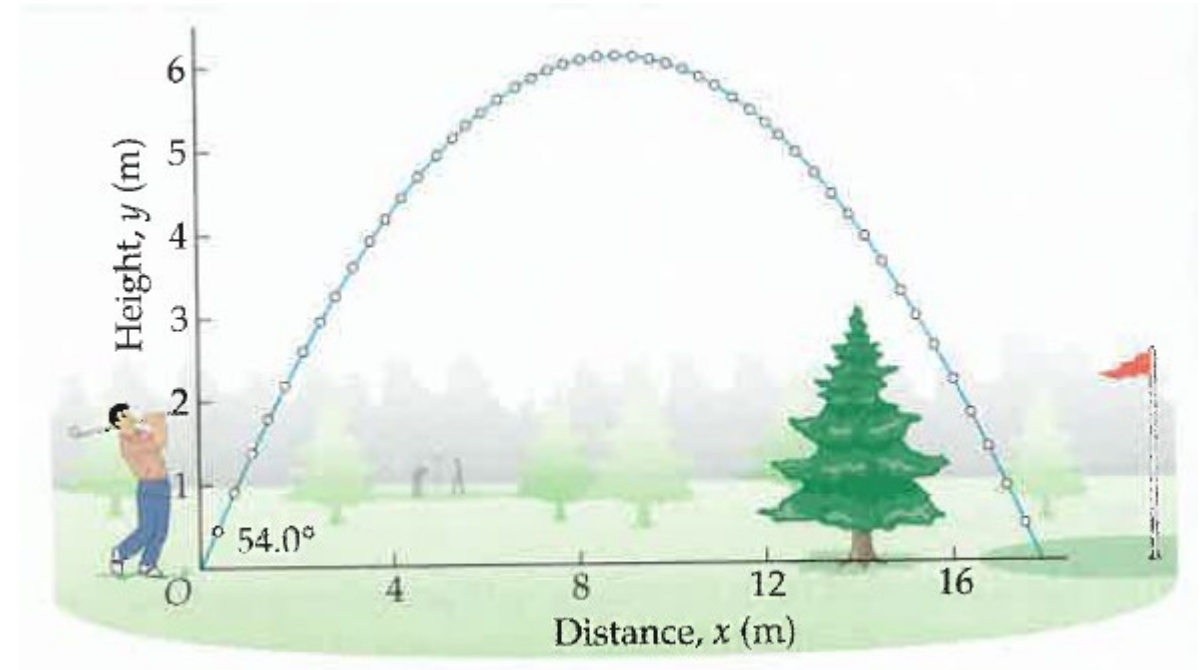
$$v_y(t) = v_{y,0} - gt = 11.47 - g \times 0.5 = 6.57 \text{ m/s}$$

$$\text{Velocity: } \vec{v}(t) = v_x \hat{i} + v_y \hat{j} = 16.38 \hat{i} + 6.57 \hat{j}$$

Case Problem 5

Chipping from the rough, a golfer sends the ball over a 3.00m-high tree that is 14.0m away. The ball lands at the same level from which it was stuck after travelling a horizontal distance of 17.8m – on the green. Take $g = 9.81 \text{ m/s}^2$.

- a) If the ball left the club 54.0° above the horizontal and landed on the green 2.24s later, what was its initial speed?
- b) How high was the ball when it passed over the tree?



J. S. Walker, Physics: Example 4-5, p90

Case Problem 5: Solution

$$\text{a) } x = x_0 + v_{x,0}t \quad \rightarrow \quad 17.8 = v_{x,0} \times 2.24 \quad \rightarrow \quad v_{x,0} = \frac{17.8}{2.24} = 7.95 \text{ m/s}$$

$$v_{x,0} = v_0 \cos \theta \quad \rightarrow \quad v_0 = \frac{v_{x,0}}{\cos \theta} = \frac{7.95}{\cos 54.0^\circ} = 13.5 \text{ m/s}$$

Case Problem 5: Solution

$$\text{b)} \quad x = x_0 + v_{x,0}t \quad \rightarrow \quad 14.0 = 7.95 \times t \quad \rightarrow \quad t = \frac{14.0}{7.95} = 1.76 \text{ s}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = (v_0 \sin \theta) t - \frac{1}{2}gt^2 = 4.03 \text{ m}$$

Case Problem 6

A golfer hits a ball from the origin with an initial speed of 30.0 m/s at an angle of 50.0° above the horizontal. The ball lands on a green that is 5.00m above the level where the ball was struck. Take $g = 9.81 \text{ m/s}^2$.

- a) How long is the ball in the air?
- b) How far has the ball traveled in the horizontal direction when it lands?
- c) What is the speed and direction of motion of the ball just before it lands?

Case Problem 6: Solution

a) $v_{y,0} = v_0 \sin \theta$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow 5.00 = (v_0 \sin \theta) t - \frac{1}{2}gt^2$$

$$\rightarrow t = 0.23s \text{ or } 4.46s$$

$$v_y(t) = v_{y,0} - gt$$

$$v_y(0.23) = v_0 \sin \theta - g \times 0.229 = 20.73 \text{ m/s}$$

$$v_y(4.46) = v_0 \sin \theta - g \times 4.46 = -20.77 \text{ m/s}$$

So $t = 4.46s$.

Case Problem 6: Solution

$$\text{b)} \quad v_{y,0} = v_0 \sin \theta$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow 5.00 = (v_0 \sin \theta) t - \frac{1}{2}gt^2$$

$$\rightarrow t = 4.46s$$

$$x = x_0 + v_{x,0}t = v_0 \cos \theta \times 4.46 = 86.0 \text{ m}$$

Case Problem 6: Solution

$$\begin{aligned} \text{c)} \quad v_{y,0} &= v_0 \sin \theta & y &= y_0 + v_{y,0}t - \frac{1}{2}gt^2 \rightarrow 5.00 = (v_0 \sin \theta) t - \frac{1}{2}gt^2 \\ & & & \rightarrow t = 4.46s \end{aligned}$$

$$v_x(t) = v_{x,0} = 19.28 \text{ m/s}$$

$$v_y(t) = v_{y,0} - gt \rightarrow v_y(4.46) = v_0 \sin \theta - g \times 4.46 = -20.77 \text{ m/s}$$

$$\text{Velocity: } \vec{v}(t) = v_x \hat{i} + v_y \hat{j} = 19.28 \hat{i} - 20.77 \hat{j}$$

$$v(4.46) = \sqrt{(19.28)^2 + (-20.77)^2} = 28.34 \text{ m/s}$$

$$\theta = \tan^{-1} \frac{v_y(4.46)}{v_x(4.46)} = \tan^{-1} \frac{-20.77}{19.28} = 360^\circ - 47.1^\circ = 312.9^\circ$$

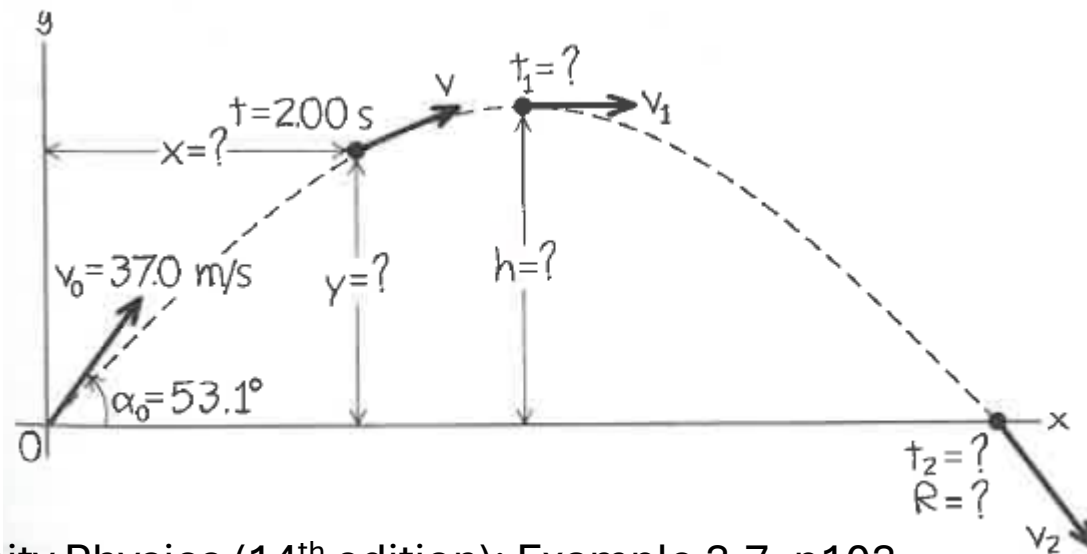
J. S. Walker, Physics: Active Example 4-1, p91

Off-class practice question(s)

Case Problem

A batter hits a baseball so that it leaves the bat at speed $v_0 = 37.0 \text{ m/s}$ at an angle $\alpha_0 = 53.1^\circ$. Take $g = 9.8 \text{ m/s}^2$.

- Find the position of the ball and its velocity (magnitude and direction) at $t=2.00 \text{ s}$.
- Find the time when the ball reaches the highest point of its flight.
- Find the horizontal range R – that is, the horizontal distance from the starting point to where the ball hits the ground – and the ball's velocity just before it hits.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.7, p103

Case Problem: Solution

a)

$$v_{x,0} = v_0 \cos \theta = 37 \times \cos 53.1^\circ = 22.2 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 37 \times \sin 53.1^\circ = 29.6 \text{ m/s}$$

$$x = x_0 + v_{x,0}t = 22.2 \times 2 = 44.4 \text{ m}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 29.6 \times 2 - \frac{1}{2}g \times (2)^2 = 39.6 \text{ m}$$

$$\vec{r}(t) = x\hat{i} + y\hat{j} = (44.4\hat{i} + 39.6\hat{j}) \text{ m}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.7, p103

Case Problem: Solution

$$\text{a)} \quad v_{x,0} = v_0 \cos \theta = 37 \times \cos 53.1^\circ = 22.2 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 37 \times \sin 53.1^\circ = 29.6 \text{ m/s}$$

$$v_x(2) = v_{x,0} = 22.2 \text{ m/s}$$

$$v_y(t) = v_{y,0} - gt \rightarrow v_y(2) = 29.6 - g \times 2 = 10 \text{ m/s}$$

$$\text{Velocity: } \vec{v}(t) = v_x \hat{i} + v_y \hat{j} = 22.2 \hat{i} + 10 \hat{j}$$

$$v(2) = \sqrt{(22.2)^2 + (10)^2} = 24.4 \text{ m/s}$$

$$\theta = \tan^{-1} \frac{v_y(2)}{v_x(2)} = \tan^{-1} \frac{10}{22.2} = 24.2^\circ$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.7, p103

Case Problem: Solution

$$\text{b)} \quad v_{x,0} = v_0 \cos \theta = 37 \times \cos 53.1^\circ = 22.2 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 37 \times \sin 53.1^\circ = 29.6 \text{ m/s}$$

$$v_y = v_{y,0} - gt = 29.6 - gt \rightarrow v_y(T) = 0 = 29.6 - 9.8T \rightarrow T = 3.02 \text{ s}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 29.6t - \frac{1}{2}gt^2 \rightarrow y(T) = 29.6T - \frac{1}{2}gT^2 = 44.7 \text{ m}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.7, p103

Case Problem: Solution

$$\text{c)} \quad v_{x,0} = v_0 \cos \theta = 37 \times \cos 53.1^\circ = 22.2 \text{ m/s}$$

$$v_{y,0} = v_0 \sin \theta = 37 \times \sin 53.1^\circ = 29.6 \text{ m/s}$$

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2 = 29.6t - \frac{1}{2}gt^2 = 0 \quad \rightarrow \quad t = 6.04 \text{ s}$$

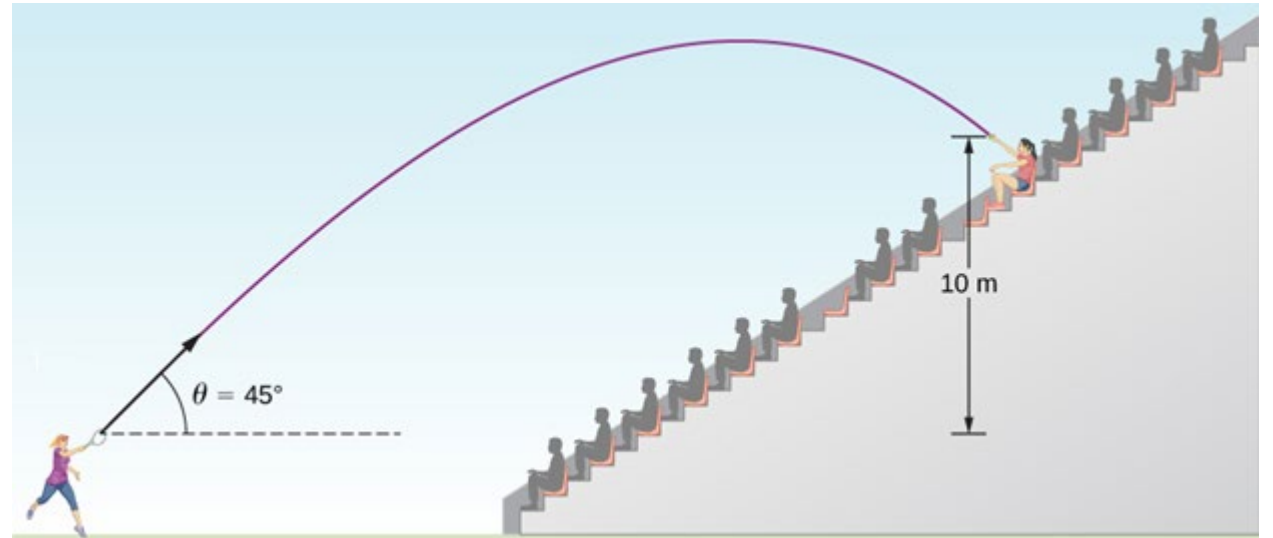
$$x = x_0 + v_{x,0}t = 22.2 \times 6.04 = 134.1 \text{ m} = R$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 3.7, p103

Case Problem

A player throws a ball with an initial speed of 25 m/s into the stands at an angle of 45° above the horizontal. The ball is caught by a person 10 m above the level at which the ball was thrown. Take $g = 9.81 \text{ m/s}^2$.

- (a) Find the time of flight when the ball reaches the person.
- (b) Find the magnitude and direction of the ball's velocity at impact.



<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem: Solution

A player throws a ball with an initial speed of 25 m/s into the stands at an angle of 45° above the horizontal. The ball is caught by a person 10 m above the level at which the ball was thrown. Take $g = 9.81 \text{ m/s}^2$.

(a) Find the time of flight when the ball reaches the person.

$$y = y_0 + v_{y,0}t - \frac{1}{2}gt^2$$

$$10 = v_0 \sin \theta t - \frac{1}{2}gt^2 \rightarrow t = 0.7 \text{ or } 2.9 \text{ s}$$

$$v_y(0.7) = v_{y,0} - g \times 0.7 = 10.8 \text{ m/s}$$

$$v_y(2.9) = v_{y,0} - g \times 2.9 = -10.8 \text{ m/s}$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>

Case Problem: Solution

A player throws a ball with an initial speed of 25 m/s into the stands at an angle of 45° above the horizontal. The ball is caught by a person 10 m above the level at which the ball was thrown.

(b) Find the magnitude and direction of the ball's velocity at impact.

$$v_x(2.9) = v_{x,0} = 17.7 \text{ m/s} \qquad v_y(2.9) = v_{y,0} - g \times 2.9 = -10.7 \text{ m/s}$$

$$\text{Velocity: } \vec{v}(t) = v_x \hat{i} + v_y \hat{j} = 17.7 \hat{i} - 10.7 \hat{j}$$

$$v(2.9) = \sqrt{(17.7)^2 + (-10.7)^2} = 20.7 \text{ m/s}$$

$$\theta = \tan^{-1} \frac{v_y(2.9)}{v_x(2.9)} = \tan^{-1} \frac{-10.7}{17.7} = 360^\circ - 31.2^\circ = 328.8^\circ$$

<https://courses.lumenlearning.com/suny-osuniversityphysics/chapter/4-3-projectile-motion/>